Notes: Merton (Econometrica, 1973)

An Intertemporal Capital Asset Pricing Model

Contents

1 Big picture

- **Question.** What should equilibrium expected returns look like when investors choose portfolios and consumption continuously over time, and when future investment opportunities are stochastic?
- **Core contribution.** Merton derives an intertemporal capital asset pricing model directly from expected-utility maximizing consumption–portfolio choice.
- **Why CAPM is incomplete.** The classical Sharpe–Lintner–Mossin CAPM is a static model. It prices only market risk. Merton shows that in an intertemporal economy, assets can also be valuable because they hedge changes in future investment opportunities.
- **Key mechanism.** If future interest rates, volatilities, or expected returns change randomly, investors care about how today's asset returns covary with those state variables. Assets that pay off when future opportunities worsen provide insurance.
- **Demand result.** Optimal risky-asset demand has two parts: a myopic mean–variance demand plus an intertemporal hedging demand.
- **Separation result.** With a constant investment opportunity set, the standard two-fund theorem and CAPM obtain. With one stochastic state variable, a three-fund theorem obtains: a riskless fund, the tangency risky fund, and a hedge fund.
- **Pricing result.** Expected excess returns compensate investors for market covariance and covariance with adverse shifts in the investment opportunity set. The security market line becomes a security market plane.
- **Important implication.** An asset with zero market beta can have an expected return above the riskless rate if it is exposed to intertemporal hedging risk.
- **Term-structure implication.** Long-term riskless bonds can earn risk premiums even without market beta because they hedge interest-rate or opportunity-set changes.
- **Empirical motivation.** The paper connects the theory to evidence that low-beta assets earn too much and high-beta assets earn too little relative to the static CAPM.
- **Takeaway.** In dynamic asset pricing, risk is not only covariance with the market portfolio. It is also covariance with state variables that determine future investment opportunities.

2 Data, setting, and measurement

2.1 Nature of the paper

- This is a theoretical paper, not an empirical paper.
- There is no firm-level dataset, no empirical regression table, and no plotted figure.
- The central objects are stochastic processes for asset returns, state variables, investor wealth, consumption, and market clearing.
- The "data" in the model are the primitives of the capital market: expected returns, variances, correlations, interest rates, and state-variable dynamics.

Interpretation: The correct reading style is therefore not "what sample is used?" but "what primitives generate the equilibrium pricing restrictions?"

2.2 Capital-market assumptions

The paper assumes a perfect and continuously trading capital market:

- all assets have limited liability;
- there are no taxes, transaction costs, or indivisibilities;
- investors are price takers;
- markets are always in equilibrium;
- borrowing and lending occur at the same instantaneous rate;
- short sales are allowed;
- trading occurs continuously over time.

Interpretation: Continuous trading is not a cosmetic assumption. It lets the model use diffusion processes and makes instantaneous means and variances sufficient for local portfolio choice.

2.3 Investment opportunity set

At each date, investors need to know:

- the current transition probabilities for asset returns;
- the stochastic process governing future changes in these transition probabilities.

Merton calls the set of current means, volatilities, correlations, and related state variables the investment opportunity set.

Interpretation: Static CAPM assumes the opportunity set is effectively fixed over the relevant horizon. Merton's model allows it to change, so investors care about hedging those changes.

2.4 Assets and firms

Each firm is treated as holding a single asset and issuing one class of equity. The paper defines:

- $K(t)$: physical capital employed by the firm;
- $P_k(t)$: price per unit of physical capital;
- $V(t) = P_k(t)K(t)$: market value of the asset;
- $N(t)$: number of shares outstanding;
- $P(t)$: price per share.

Interpretation: This construction separates changes in per-share returns from changes in the number of shares supplied. The asset-pricing model focuses on return dynamics; a full general equilibrium model would also specify the supply side.

2.5 Riskless asset

The model contains n risky assets and one instantaneous riskless asset. "Instantaneously riskless" means that at each instant investors know the return over the next instant:

$$\alpha_{n+1} = r(t), \qquad \sigma_{n+1} = 0.$$

Future values of $r(t)$ can still be stochastic.

Interpretation: A short bond can be riskless instantaneously while the future short rate is risky. This distinction is crucial for the intertemporal hedging term.

3 Models and theoretical specifications

3.1 Return dynamics

For risky asset i:

$$\frac{dP_i}{P_i} = \alpha_i \, dt + \sigma_i \, dz_i, \tag{1}$$

where dz_i is a Brownian increment, α_i is the instantaneous expected return, and σ_i^2 is the instantaneous variance.

The opportunity set itself can change:

$$d\alpha_i = a_i \, dt + b_i \, dq_i, \qquad d\sigma_i = f_i \, dt + g_i \, d\chi_i. \tag{2}$$

Interpretation: The paper prices assets in a world where return distributions are locally diffusion-like but their parameters can move stochastically over time.

3.2 Investor objective

Investor k chooses consumption and portfolio weights to maximize expected lifetime utility:

$$\max E_0 \left[\int_0^{T_k} U_k(c_k(s), s) \, ds + B_k(W_k(T_k), T_k) \right]. \tag{3}$$

Interpretation: This is the dynamic replacement for one-period mean–variance choice. The investor values both current consumption and future wealth.

3.3 Wealth dynamics

Let w_i be the share of wealth invested in risky asset i. The riskless asset weight adjusts so that portfolio weights sum to one. Suppressing wage income, wealth evolves as

$$dW = \left[\sum_{i=1}^{n} w_i(\alpha_i - r) + r \right] W \, dt + \sum_{i=1}^{n} w_i W \sigma_i \, dz_i - c \, dt. \tag{4}$$

Interpretation: Portfolio choice affects both drift and risk in wealth. Consumption reduces wealth directly.

3.4 State-variable dynamics

Let X collect the state variables that determine the opportunity set. Merton writes

$$dX = F(X) \, dt + G(X) \, dQ. \tag{5}$$

Interpretation: The state variables summarize everything investors need to forecast future opportunities. This is the state-space version of intertemporal asset pricing.

3.5 Dynamic programming equation

Let $J(W, t, X)$ be the derived utility of wealth. The Bellman equation is:

$$
\begin{aligned}
0 = \max_{\{c,w\}} \{ &U(c,t) + J_t + J_W \left(\left[\sum_i w_i(\alpha_i - r) + r \right] W - c \right) + \sum_j J_j f_j \\
&+ \frac{1}{2} J_{WW} \sum_i \sum_j w_i w_j \sigma_{ij} W^2 + \sum_i \sum_j J_{Wj} w_i W \sigma_i g_j \eta_{ij} + \frac{1}{2} \sum_i \sum_j J_{ij} g_i g_j \nu_{ij} \}.
\end{aligned}
\tag{6}
$$

Interpretation: The Bellman equation includes the effect of portfolio returns on wealth and the joint movement of returns with state variables.

3.6 First-order conditions

The consumption condition is

$$U_c(c,t) = J_W(W,t,X). \tag{7}$$

The portfolio first-order condition is

$$0 = J_W(\alpha_i - r) + J_{WW} \sum_j w_j W \sigma_{ij} + \sum_j J_{Wj} g_j \sigma_i \eta_{ij}. \tag{8}$$

Interpretation: The first term is expected excess return. The second is the standard risk penalty. The third is the intertemporal hedging term.

3.7 Asset-demand equation

Solving the linear first-order conditions gives

$$w_i W = A \sum_j v_{ij}(\alpha_j - r) + \sum_k \sum_j H_k \sigma_j g_k \eta_{jk} v_{ij}, \tag{9}$$

where $[v_{ij}]$ is the inverse variance–covariance matrix,

$$A = -\frac{J_W}{J_{WW}} > 0, \qquad H_k = -\frac{J_{Wk}}{J_{WW}}.$$

Interpretation: The first term is myopic mean–variance demand. The second term is hedging demand against shifts in the opportunity set.

3.8 Constant opportunity set and classical CAPM

If α, σ, correlations, and r are constant, the hedging term disappears:

$$w_i W = A \sum_j v_{ij}(\alpha_j - r). \tag{10}$$

Then the classical two-fund theorem obtains, and equilibrium expected returns satisfy

$$\alpha_i - r = \beta_i(\alpha_M - r), \qquad \beta_i = \frac{\sigma_{iM}}{\sigma_M^2}. \tag{11}$$

Interpretation: The classical CAPM is a special case of Merton's model. It requires away stochastic future investment opportunities.

3.9 One state variable and the three-fund theorem

Merton specializes to the case where one state variable—the short interest rate r—summarizes shifts in the opportunity set. Asset demand becomes

$$d_i^k = A^k \sum_j v_{ij}(\alpha_j - r) + H^k \sum_j v_{ij}\sigma_{jr}. \tag{12}$$

If asset n is perfectly negatively correlated with changes in r, then the demand function simplifies and investors can be served by three funds:

- a risky tangency fund;
- a hedge fund, interpreted conceptually as a long-term bond;
- the instantaneous riskless asset.

Interpretation: The hedge fund is not there to improve the static mean–variance frontier. It is there to insure against bad shifts in the future opportunity set.

3.10 Equilibrium yield relationship

Aggregating demands and imposing market clearing gives:

$$\alpha_i - r = \frac{M}{A}\sigma_{iM} + \frac{Hg}{A\sigma_n}\sigma_{in}. \tag{13}$$

Merton rewrites this as a security market plane:

$$\alpha_i - r = \frac{\sigma_i(\rho_{iM} - \rho_{in}\rho_{nM})}{\sigma_M(1-\rho_{nM}^2)}(\alpha_M - r) + \frac{\sigma_i(\rho_{in} - \rho_{iM}\rho_{nM})}{\sigma_n(1-\rho_{nM}^2)}(\alpha_n - r). \tag{14}$$

Interpretation: Expected returns depend on market risk and hedge-state risk. The static security market line is replaced by a two-factor pricing plane.

3.11 When does the model collapse to CAPM?

The model reduces to the classical CAPM if the aggregate hedging demand term disappears:

$$H = 0,$$

or if all asset returns are uncorrelated with changes in the relevant state variable:

$$\sigma_{ir} = 0.$$

Interpretation: These are restrictive conditions. Merton's point is that CAPM is not generally valid once future opportunities are stochastic.

4 Conceptual design and empirical implications

4.1 Why the model is a genuine equilibrium theory

- The paper starts with individual utility maximization.
- It derives individual asset demands.
- It aggregates demands across investors.
- It imposes market clearing.
- It then obtains restrictions on equilibrium expected returns.

Interpretation: This is not just a reduced-form factor model. The factor structure comes from hedging motives in a dynamic optimization problem.

4.2 Why a zero-beta asset can earn a premium

In the static CAPM, if $\beta_i = 0$, expected excess return should be zero. In Merton's model, an asset can have zero market beta but still be correlated with adverse shifts in the opportunity set. Then it can require, or provide, a risk premium.

Interpretation: This is one of the paper's most important departures from classical CAPM: market beta is not the only systematic risk.

4.3 Connection to the term structure

A long-term default-free bond may be nearly uncorrelated with the market portfolio but strongly correlated with changes in the short rate. Such a bond can serve as a hedge asset.

Interpretation: Term premiums need not be "liquidity premiums." They can reflect compensation for bearing or providing intertemporal hedging services.

4.4 Connection to Black–Jensen–Scholes evidence

The paper discusses evidence that low-beta assets earned higher returns and high-beta assets lower returns than static CAPM predicts. Merton links this to a second systematic factor:

$$\alpha_i - r = \beta_i(\alpha_M - r) + \gamma_i(\alpha_0 - r).$$

Interpretation: Merton does not provide a full empirical test. The point is that the ICAPM gives a theoretical reason to expect additional priced risks beyond market beta.

4.5 Limits of the paper

- The model keeps many perfect-market assumptions: no taxes, no transaction costs, no short-sale restrictions, and homogeneous expectations.
- The supply side of firms is not fully modeled.
- The key empirical state variables are not fully identified.

- The one-state-variable version uses the interest rate as an instrumental representation of a broader opportunity set.
- The empirical discussion is preliminary.

Interpretation: The paper is foundational because it supplies the logic of intertemporal hedging demand. It does not settle which state variables matter most empirically.

5 Tables, figures, and original equation panels

The paper has no empirical tables or plotted figures. To preserve the requested format, this section directly extracts and groups the original equation/theorem panels from the paper.

5.1 Panel 1: Return dynamics and investor objective

independent.[15] If we define the stochastic process, $z(t)$, by

$$z(t+h) = z(t) + y(t)\sqrt{h}, \tag{4}$$

then $z(t)$ is a stochastic process with independent increments. If it is further assumed that $y(t)$ is Gaussian distributed,[16] then the limit as h tends to zero of $z(t+h) - z(t)$ describes a Wiener process or Brownian motion. In the formalism of stochastic differential equations,

$$dz \equiv y(t)\sqrt{dt}. \tag{5}$$

In a similar fashion, we can take the limit of (3) to derive the stochastic differential equation for the instantaneous return on the ith asset as

$$\frac{dP_i}{P_i} = \alpha_i\,dt + \sigma_i\,dz_i. \tag{6}$$

Processes such as (6) are called Itô processes and while they are continuous, they are not differentiable.[17]

From (6), a sufficient set of statistics for the opportunity set at a given point in time is $\{\alpha_i, \sigma_i, \rho_{ij}\}$ where ρ_{ij} is the instantaneous correlation coefficient between the Wiener processes dz_i and dz_j. The vector of return dynamics as described in (6) will be Markov only if $\alpha_i.\,\sigma_i$, and ρ_{ij} were, at most, functions of the P's. In general, one would not expect this to be the case since, at each point in time, equilibrium clearing conditions will define a set of implicit functions between equilibrium market values, $V_i(t) = N_i(t)P_i(t)$, and the α_i, σ_i, and ρ_{ij}. Hence, one would expect the changes in required expected returns to be stochastically related to changes in market values, and dependence on P solely would obtain only if changes in N (changes in supplies) were non-stochastic. Therefore, to close the system, we append the dynamics for the changes in the opportunity set over time: namely,

$$\begin{aligned} d\alpha_i &= a_i\,dt + b_i\,dq_i, \\ d\sigma_i &= f_i\,dt + g_i\,dx_i, \end{aligned} \tag{7}$$

where we do assume that (6) and (7), together, form a Markov system,[18] with dq_i and dx_i standard Wiener processes.

[15] It is sufficient to assume that the $y(t)$ are uncorrelated and that the higher order moments are

(a) Return and opportunity-set dynamics.

Under the assumptions of continuous trading and the continuous Markov structure of the stochastic processes, it has been shown that the instantaneous, first two moments of the distributions are sufficient statistics.[19] Further, by the existence and boundedness of α and σ, P equal to zero is a natural absorbing barrier ensuring limited liability of all assets.

For the rest of the paper, it is assumed that there are n distinct[20] risky assets and one "instantaneously risk-less" asset. "Instantaneously risk-less" means that, at each instant of time, each investor knows with certainty that he can earn rate of return $r(t)$ over the next instant by holding the asset (i.e., $\sigma_{n+1} = 0$ and $\alpha_{n+1} \equiv r(t)$). However, the future values of $r(t)$ are not known with certainty (i.e., $b_{n+1} \neq 0$ in (7)). We interpret this asset as the exchange asset and $r(t)$ as the instantaneous private sector borrowing (and lending) rate. Alternatively, the asset could represent (very) short government bonds.

4. PREFERENCE STRUCTURE AND BUDGET EQUATION DYNAMICS

We assume that there are K consumer-investors with preference structures as described in [**25**]: namely, the kth consumer acts so as to

$$\max E_0\left[\int_0^{T^k} U^k[c^k(s), s]\,ds + B^k[W^k(T^k), T^k]\right], \tag{8}$$

where "E_0" is the conditional expectation operator, conditional on the current value of his wealth, $W^k(0) = W^k$ are the state variables of the investment opportunity set, and T^k is the distribution for his age of death (which is assumed to be independent of investment outcomes). His instantaneous consumption flow at age t is $c^k(t)$.[21] U^k is a strictly concave von Neumann-Morgenstern utility function for consumption and B^k is a strictly concave "bequest" or utility-of-terminal wealth function.

Dropping the superscripts (except where required for clarity), we can write the accumulation equation for the kth investor as[22]

$$dW = \sum_1^{n+1} w_i W\,dP_i/P_i + (y - c)\,dt, \tag{9}$$

(b) Investor objective and wealth budget.

Read:

- The asset return is modeled as an Ito process.
- The expected return and volatility of each asset can themselves move over time.
- The investor maximizes expected utility from consumption plus terminal wealth.
- Wealth changes through portfolio returns and consumption.

Economic message: These panels set up the dynamic problem: portfolio choice is not just about today's return distribution but also about future changes in the opportunity set.

5.2 Panel 2: Bellman equation and optimal demand

for dP_i/P_i from (6), we can re-write (9) as

$$dW = \left[\sum_1^n w_i(\alpha_i - r) + r\right] W\,dt + \sum_1^n w_i W \sigma_i\,dz_i + (y - c)\,dt, \tag{10}$$

where his choice for $w_1, w_2, \ldots, w_n$ is unconstrained because w_{n+1} can always be chosen to satisfy the budget constraint $\Sigma_1^{n+1} w_i = 1$.

From the budget constraint, $W = \Sigma_1^{n+1} N_i P_i$, and the accumulation equation (9), we have that

$$(y - c)\,dt = \sum_1^{n+1} dN_i(P_i + dP_i), \tag{11}$$

i.e., the net value of new shares purchased must equal the value of savings from wage income.

5. THE EQUATIONS OF OPTIMALITY: THE DEMAND FUNCTIONS FOR ASSETS

For computational simplicity, we will assume that investors derive all their income from capital gains sources (i.e., $y \equiv 0$),[23] and for notational simplicity, we introduce the state-variable vector, X, whose m elements, x_i, denote the current levels of P, α, and σ. The dynamics for X are written as the vector Itô process,

$$dX = F(X)\,dt + G(X)\,dQ, \tag{12}$$

where F is the vector $[f_1, f_2, \ldots, f_m]$, G is a diagonal matrix with diagonal elements $[g_1, g_2, \ldots, g_m]$, dQ is the vector Wiener process $[dq_1, dq_2, \ldots, dq_m]$, η_{ij} is the instantaneous correlation coefficient between dq_i and dz_j, and v_{ij} is the instantaneous correlation coefficient between dq_i and dq_j.

I have shown elsewhere[24] that the necessary optimality conditions for an investor who acts according to (8) in choosing his consumption-investment program are that, at each point in time,

$$\begin{aligned} 0 = \max_{(c,w)} \Bigg[& U(c,t) + J_t + J_W\left[\left(\sum_1^n w_i(\alpha_i - r) + r\right)W - c\right] \\ & + \sum_1^m J_i f_i + \tfrac{1}{2} J_{WW} \sum_1^n \sum_1^n w_i w_j \sigma_{ij} W^2 \\ & + \sum_1^m \sum_1^n J_{iW} w_j W g_i \sigma_j \eta_{ij} + \frac{1}{2} \sum_1^m \sum_1^m J_{ij} g_i g_j v_{ij} \Bigg], \end{aligned} \tag{13}$$

(a) Dynamic programming equation.

subject to $J(W, T, X) = B(W, T)$, where subscripts on the "derived" utility of wealth function, J, denote partial derivatives. The σ_{ij} are the instantaneous covariances between the returns on the ith and jth assets ($\equiv \sigma_i \sigma_j \rho_{ij}$).

The $n + 1$ first-order conditions derived from (13) are

$$0 = U_c(c, t) - J_W(W, t, X), \tag{14}$$

and

$$0 = J_W(\alpha_i - r) + J_{WW} \sum_1^n w_j W \sigma_{ij} + \sum_1^m J_{jW} g_j \sigma_i \eta_{ji} \qquad (i = 1, 2, \ldots, n), \tag{15}$$

where $c = c(W, t, X)$ and $w_i = w_i(W, t, X)$ are the optimum consumption and portfolio rules as functions of the state variables. Equation (14) is the usual intertemporal envelope condition to equate the marginal utility of current consumption to the marginal utility of wealth (future consumption). The manifest characteristic of (15) is its linearity in the portfolio demands; hence, we can solve explicitly for these functions by matrix inversion,

$$w_i W = A \sum_1^n v_{ij}(\alpha_j - r) + \sum_1^m \sum_1^n H_k \sigma_j g_k \eta_{jk} v_{ij} \qquad (i = 1, 2, \ldots, n), \tag{16}$$

where the v_{ij} are the elements of the inverse of the instantaneous variance-covariance matrix of returns, $\Omega = [\sigma_{ij}]$, $A \equiv -J_W/J_{WW}$, and $H_k \equiv -J_{kW}/J_{WW}$.

Some insight in interpreting (16) can be gained by expressing A and H_k in terms of the utility and consumption functions: namely, by the implicit function theorem applied to (14),

$$A = -U_c \bigg/ \left(U_{cc} \frac{\partial c}{\partial W}\right) > 0, \tag{17}$$

and

$$H_k = -\frac{\partial c}{\partial x_k} \bigg/ \frac{\partial c}{\partial W} \gtreqless 0. \tag{18}$$

(b) First-order conditions and demand decomposition.

Read:

- The HJB equation includes wealth risk and state-variable risk.
- The consumption condition equates marginal utility of consumption with marginal value of wealth.
- Portfolio demand is linear in expected excess returns and in covariances with state variables.
- Equation (16) is the central demand equation of the paper.

Economic message: The first part of demand is myopic. The second is intertemporal hedging demand. This is the core innovation.

5.3 Panel 3: CAPM as a special case

(homogeneous expectations), then the ratio of the demands for risky assets will be independent of preferences, and the same for all investors. Further, we have the following theorem.

THEOREM 1:[28] *Given n risky assets whose returns are log-normally distributed and a riskless asset, then (i) there exists a unique pair of efficient portfolios ("mutual funds") one containing only the riskless asset and the other only risky assets, such that, independent of preferences, wealth distribution, or time horizon, all investors will be indifferent between choosing portfolios from among the original $n + 1$ assets or from these two funds; (ii) the distribution of the return on the risky fund is log-normal; (iii) the proportion of the risky fund's assets invested in the kth asset is*

$$\sum_1^n v_{kj}(\alpha_j - r) \Big/ \sum_1^n \sum_1^n v_{ij}(\alpha_i - r) \qquad (k = 1, 2, \dots, n).$$

Theorem 1 is the continuous-time version of the Markowitz-Tobin separation theorem and the holdings of the risky fund correspond to the optimal combination of risky assets (see Sharpe [**39**, p. 69]).

Using the condition that the market portfolio is efficient in equilibrium, it can be shown (see Merton [**26**]) that, for this version of the model, the equilibrium returns will satisfy

$$\alpha_i - r = \beta_i(\alpha_M - r) \qquad (i = 1, 2, \dots, n), \tag{20}$$

where $\beta_i \equiv \sigma_{iM}/\sigma_M^2$, σ_{iM} is the covariance of the return on the ith asset with the return on the market portfolio, and α_M is the expected return on the market portfolio. Equation (20) is the continuous-time analog to the security market line of the classical capital asset pricing model.

Hence, the additional assumption of a constant investment opportunity set is a sufficient condition for investors to behave as if they were single-period maximizers and for the equilibrium return relationship specified by the capital asset pricing model to obtain. Except for some singular cases, this assumption is also

Figure 3: Constant opportunity set, two-fund separation, and the CAPM security market line.

Read:

- If the investment opportunity set is constant, the hedging term disappears.
- Theorem 1 gives the continuous-time two-fund theorem.
- Equation (20) gives the classical CAPM relation.

Economic message: Classical CAPM is valid only under special assumptions. Merton's model nests it but does not generally imply it.

5.4 Panel 4: Interest-rate state variable and three-fund theorem

changing stochastically over time. The simplest form of the model consistent with this observation occurs if it is assumed that a single state variable is sufficient to describe changes in the opportunity set. We further assume that this variable is the interest rate (i.e., $\alpha_i = \alpha_i(r)$ and $\sigma_i = \sigma_i(r)$).

The interest rate has always been an important variable in portfolio theory, general capital theory, and to practitioners. It is observable, satisfies the condition of being stochastic over time, and while it is surely not the sole determinant of yields on other assets,[28] it is an important factor. Hence, one should interpret the effects of a changing interest rate in the forthcoming analysis in the way economists have generally done in the past: namely, as a single (instrumental) variable representation of shifts in the investment opportunity set. For example, $\partial c/\partial r$ is the change in consumption due to a change in the opportunity set for a fixed level of wealth.

This assumed, we can write the kth investor's demand function for the ith asset, (16), as

$$d_i^k = A^k \sum_1^n v_{ij}(\alpha_j - r) + H^k \sum_1^n v_{ij}\sigma_{jr} \qquad (i = 1, 2, \ldots, n), \tag{21}$$

where $d_i^k \equiv w_i^k W^k$; $H^k \equiv -(\partial c^k/\partial r)/(\partial c^k/\partial W^k)$, and σ_{jr} is the (instantaneous) covariance between the return on the jth asset and changes in the interest rate ($\equiv \rho_{jr}\sigma_j g$). By inspection of (21), the ratio of the demands for risky assets is a function of preferences, and hence, the standard separation theorem does not obtain. However, generalized separation (see [**5**]) does obtain. In particular, it will be shown that all investors' optimal portfolios can be represented as a linear combination of three mutual funds (portfolios).

Although not necessary for the theorem, it will throw light on the analysis to assume there exists an asset (by convention, the nth one) whose return is perfectly negatively correlated with changes in r, i.e., $\rho_{nr} = -1$. One such asset might be riskless (in terms of default), long-term bonds.[30] In this case, we can re-write the covariance term σ_{jr} as

$$\sigma_{jr} = \rho_{jr}\sigma_j g, \tag{22}$$

(a) Demand when the short rate represents the state variable.

re-write (21) in the simplified form,

$$d_i^k = A^k \sum_1^n v_{ij}(\alpha_j - r) \qquad (i = 1, 2, \ldots, n-1), \tag{23}$$

$$d_n^k = A^k \sum_1^n v_{nj}(\alpha_j - r) - gH^k/\sigma_n.$$

THEOREM 2 ("*Three Fund*" *Theorem*): *Given n risky assets and a riskless asset satisfying the conditions of this section, then there exist three portfolios ("mutual funds") constructed from these assets, such that* (i) *all risk-averse investors, who behave according to* (8), *will be indifferent between choosing portfolios from among the original $n + 1$ assets or from these three funds*; (ii) *the proportions of each fund's portfolio invested in the individual assets are purely "technological"* (*i.e., depend only on the variables in the investment opportunity set for individual assets and not on investor preferences*); *and* (iii) *the investor's demands for the funds do not require knowledge of the investment opportunity set for the individual assets nor of the asset proportions held by the funds.*

PROOF: Let the first fund hold the same proportions as the risky fund in Theorem 1: namely, $\delta_k = \Sigma_1^n v_{kj}(\alpha_j - r)/\Sigma_1^n\Sigma_1^n v_{ij}(\alpha_j - r)$, for $k = 1, 2, \ldots, n$. Let the second fund hold only the nth asset and the third fund only the riskless asset. Let λ_i^k be the fraction of the kth investor's wealth invested in the ith fund, $i = 1, 2, 3$ ($\Sigma_1^3 \lambda_i^k = 1$). To prove (i), we must show that there exists an allocation $(\lambda_1^k, \lambda_2^k)$ which exactly replicates the demand functions, (23), i.e., that

$$\lambda_1^k \delta_i = (A^k/W^k) \sum_1^n v_{ij}(\alpha_j - r) \qquad (i = 1, 2, \ldots, n-1), \tag{24}$$

$$\lambda_1^k \delta_n + \lambda_2^k = (A^k/W^k) \sum_1^n v_{nj}(\alpha_j - r) - gH^k/\sigma_n W^k.$$

From the definition of δ_i, the allocation $\lambda_1^k = (A^k/W^k)\Sigma_1^n\Sigma_1^n v_{ij}(\alpha_j - r)$ and $\lambda_2^k = -gH^k/\sigma_n W^k$ satisfied (24). Part (ii) follows from the choice for the three funds. To prove (iii), we must show that investors will select this allocation, given only the knowledge of the (aggregated) investment opportunity set, i.e., given $(\alpha, \alpha_n, r, \sigma, \sigma_n, \rho, g)$ where α and σ^2 is the expected return and variance on the first fund's portfolio and ρ is its covariance with the return on the second fund. From the definition of δ_i, it is straightforward to show that $(\alpha - r)/\sigma^2 = \Sigma_1^n\Sigma_1^n v_{ij}(\alpha_i - r)$ and $\rho = \sigma(\alpha_n - r)/\sigma_n(\alpha - r)$. The demand functions for the funds

(b) Three-fund theorem and proof sketch.

Read:

- The interest rate is treated as a compact proxy for stochastic investment opportunities.
- If one asset hedges interest-rate changes, the hedging component can be separated into one hedge fund.
- The three funds are the risky fund, the hedge fund, and the riskless asset.

Economic message: Dynamic asset pricing creates a demand for a hedge portfolio. This is why a three-fund theorem replaces the two-fund theorem.

5.5 Panel 5: Equilibrium expected returns

the stochastic differential equations

$$N\,dP_M \equiv \sum_1^{n+1} N_i\,dP_i, \tag{28}$$

$$dN(P_M + dP_M) \equiv \sum_1^{n+1} dN_i(P_i + dP_i),$$

where, by construction, dP_M/P_M is the rate of return on the market (portfolio).

Substituting from (27) into (28) and using (11), we have

$$dN(P_M + dP_M) = \sum_1^{K} (y^i - c^i)\,dt, \tag{29}$$

$$N\,dP_M = \sum_1^{n+1} D_i\,dP_i/P_i.$$

If $w_i \equiv N_iP_i/M = D_i/M$, the percentage contribution of the ith firm to total market value, then, from (6) and (29), the rate of return on the market can be written as

$$\frac{dP_M}{P_M} = \left[\sum_1^n w_j(\alpha_j - r) + r\right] dt + \sum_1^n w_j\sigma_j\,dz_j. \tag{30}$$

Substituting w_iM for D_i in (25), we can solve for the equilibrium expected returns on the individual assets:

$$\alpha_i - r = (M/A)\sum_1^n w_j\sigma_{ij} + (Hg/A\sigma_n)\sigma_{in} \qquad (i = 1, 2, \ldots, n). \tag{31}$$

As with any asset, we can define $\alpha_M(\equiv \Sigma_1^n w_j(\alpha_j - r) + r)$, $\sigma_{iM}(\equiv \Sigma_1^n w_j\sigma_{ij})$, and $\sigma_M^2(\equiv \Sigma_1^n w_j\sigma_{jM})$ as the (instantaneous) expected return, covariance, and variance of the market portfolio. Then (31) can be re-written as

$$\alpha_i - r = (M/A)\sigma_{iM} + (Hg/A\sigma_n)\sigma_{in} \qquad (i = 1, 2, \ldots, n), \tag{32}$$

and multiplying (32) by w_i and summing, we have

$$\alpha_M - r = (M/A)\sigma_M^2 + (Hg/A\sigma_n)\sigma_{Mn}. \tag{33}$$

Noting that the nth asset satisfies (32), we can use it together with (33) to re-write (32) as

$$\alpha_i - r = \frac{\sigma_i[\rho_{iM} - \rho_{in}\rho_{nM}]}{\sigma_M(1 - \rho_{nM}^2)}(\alpha_M - r) + \frac{\sigma_i[\rho_{in} - \rho_{iM}\rho_{nM}]}{\sigma_n(1 - \rho_{Mn}^2)}(\alpha_n - r) \tag{34}$$

(a) Market clearing and the security market plane.

(i.e., $\beta_i = 0 = \rho_{iM}$), its expected return will not be equal to the riskless rate as forecast by the usual model.

Under what conditions will the security market plane equation (34) reduce to the (continuous-time) classical security market line, equation (20)? From inspection of the demand equations (21), appropriately aggregated, the conditions are

$$H = \sum_1^K -(\partial c^k/\partial r)/(\partial c^k/\partial W^k) \equiv 0 \tag{35a}$$

or

$$\sigma_{ir} \equiv 0 \qquad (i = 1, 2, \ldots, n). \tag{35b}$$

There is no obvious reason to believe that (35a) should hold unless $\partial c^k/\partial r \equiv 0$ for each investor, and the only additive utility function for which this is so is the Bernoulli logarithmic one.[32] Condition (35b) could obtain in two ways: $g \equiv 0$, i.e., the interest rate is non-stochastic, which is not so; or $\rho_{ir} \equiv 0$, i.e., all assets' returns are uncorrelated with changes in the interest rate. While this condition is possible, it would not be a true equilibrium state.

Suppose that by a quirk of nature, $\rho_{ir} \equiv 0$ for all available real assets. Then, since the nth asset does not exist, (34) reduces to (20). Consider constructing a "man-made" security (e.g., a long-term bond) which is perfectly negatively cor-

(b) When the model reduces to the classical CAPM.

Read:

- Market clearing transforms individual hedging demands into equilibrium expected-return restrictions.
- Equation (34) prices both market risk and state-variable risk.
- The classical security market line obtains only if aggregate hedging demand is zero or state-variable covariances are irrelevant.

Economic message: Expected returns are compensation for bearing aggregate wealth risk and intertemporal opportunity-set risk.

5.6 Panel 6: Empirical interpretation

high-beta ($\beta > 1$) stocks had negative correlation and low-beta ($\beta < 1$) stocks had positive correlation. We can summarize the **BJS** specification and empirical findings as follows:

$$\alpha_i - r = \beta_i(\alpha_M - r) + \gamma_i(\alpha_0 - r), \tag{36}$$

where α_0 is the expected return on the "zero-beta" portfolio, and

$$\alpha_0 > r; \tag{37a}$$

$$\gamma_i = \gamma_i(\beta_i) \quad \text{with} \quad \gamma_i(1) = 0, \quad \text{and} \quad \partial\gamma_i/\partial\beta_i < 0. \tag{37b}$$

While the finding of a second factor is consistent with the a priori specification of our model, it cannot be said that their specific findings are in agreement with the model without some further specification of the effect of a shift in r on the investment opportunity set. However, if a shift in r is an instrumental variable for a shift in capitalization rates generally, then an argument can be made that the two are in agreement.

The plan is to show that qualitative characteristics of the coefficient $(\rho_{in} - \rho_{iM}\rho_{nM})\sigma_i/\sigma_n(1 - \rho_{nM}^2)$ in (34) as a function of β_i would be the same as γ_i in (37b), and that the empirical characteristics of the zero-beta portfolio are similar to those of a portfolio of long term bonds.

If we take the classical security market line, $\alpha_i = r + \beta_i\lambda$, where $\lambda \equiv (\alpha_M - r)$, as a reasonable approximation to the relationship among capitalization rates, α_i, then we can compute the logarithmic elasticity of α_i with respect to r as a function of β_i, to be

$$\psi(\beta_i) \equiv r(1 + \beta_i\lambda')/(r + \beta_i\lambda), \tag{38}$$

where $\lambda' \equiv \partial\lambda/\partial r$, the change in the slope of the security market line with a change in r. From (27) we have that this elasticity is almost certainly a monotone decreasing function of β_i since $\psi'(\beta_i) \equiv \partial\psi/\partial\beta_i < 0$ if $\psi(1) < 1$.[33]

If we write the value of firm i as $V_i \equiv \bar{X}_i/\alpha_i$ where $\bar{X}_i$ is the "long-run" expected earnings and α_i, the rate at which they are capitalized, then the percentage change in firm value due to a change in r can be written as

$$\left(\frac{dV_i}{V_i}\right)_- \equiv \left[\frac{\partial\bar{X}_i}{\partial r}\Big/\bar{X}_i - \frac{\partial\alpha_i}{\partial r}\Big/\alpha_i\right] dr. \tag{39}$$

(a) Connection to zero-beta evidence.

where $\phi(\beta_i) \equiv \psi(\beta_i) - \beta_i\psi(1)$ satisfies $\phi(1) = 0$ and $\phi'(\beta_i) < 0$. From (40), the correlation coefficient between $d\varepsilon$ and dr, $\rho_{\varepsilon r}$, will satisfy

$$\rho_{\varepsilon r} \gtreqless 0 \quad \text{as} \quad \beta_i \lesseqgtr 1. \tag{41}$$

From the definition of $d\varepsilon$ in (40), $\rho_{\varepsilon r}$ is the partial correlation coefficient, $\rho_{ir} - \rho_{iM}\rho_{rM}$. By definition the nth asset in (34) is perfectly negatively correlated with changes in r. Hence (41) can be rewritten as

$$\rho_{in} - \rho_{iM}\rho_{nM} \lesseqgtr 0 \quad \text{as} \quad \beta_i \lesseqgtr 1. \tag{42}$$

Hence the coefficient of $(\alpha_n - r)$ in (34) could be expected to have the same properties as γ_i in (36) and (37b).

(b) Partial correlation logic.

Read:

- Merton connects the model to the empirical finding that zero-beta portfolios earned more than the riskless rate.
- The added factor can be interpreted as a hedge-state factor, not simply a market-beta factor.
- The discussion is preliminary but points toward later multifactor asset pricing.

Economic message: The ICAPM provides a theory-based reason for priced factors beyond the market portfolio.

6 Short summary

- The paper develops a continuous-time intertemporal version of asset pricing.
- Investors maximize expected utility of lifetime consumption and terminal wealth.
- Asset returns and the investment opportunity set follow diffusion processes.
- The opportunity set may change through stochastic expected returns, volatilities, correlations, or the interest rate.

- Optimal demand contains a myopic mean–variance component and a hedging component.
- With constant opportunities, the hedging term is zero and the classical CAPM obtains.
- With one stochastic state variable, three-fund separation obtains.
- The third fund is a hedge portfolio against changes in future investment opportunities.
- In equilibrium, expected returns depend on both market covariance and state-variable covariance.
- A zero-market-beta asset can have nonzero expected excess return.
- Long-term riskless bonds can have term premiums because they hedge or expose investors to state-variable risk.
- The model helps explain why the empirical security market line may be flatter than static CAPM predicts.
- The paper is a foundation for later intertemporal and multifactor asset-pricing models.

7 Conclusion

7.1 Core conclusion

Merton shows that the classical CAPM is a special case of a broader intertemporal asset-pricing model. When future investment opportunities are stochastic, investors care about assets' covariances with state variables, not only their covariance with the market portfolio.

7.2 Economic intuition

Investors choose portfolios to finance consumption over time. If future opportunities can deteriorate, they value assets that pay off in those bad opportunity-set states. Such assets may have low or zero market beta, but they still affect lifetime utility. Equilibrium expected returns must therefore compensate for both market risk and intertemporal hedging risk.

7.3 Takeaway

The paper's enduring message is that dynamic asset pricing is inherently multifactor. The market portfolio is not enough whenever the opportunity set changes over time. Intertemporal hedging demand is the bridge from CAPM to modern state-variable and factor-based asset pricing.